10.2.3. Digital Multimeter

It is a type of meter which shows measurement in digits, is called as digital multimeter. Figure 10.14 shows a typical digital multimeter.

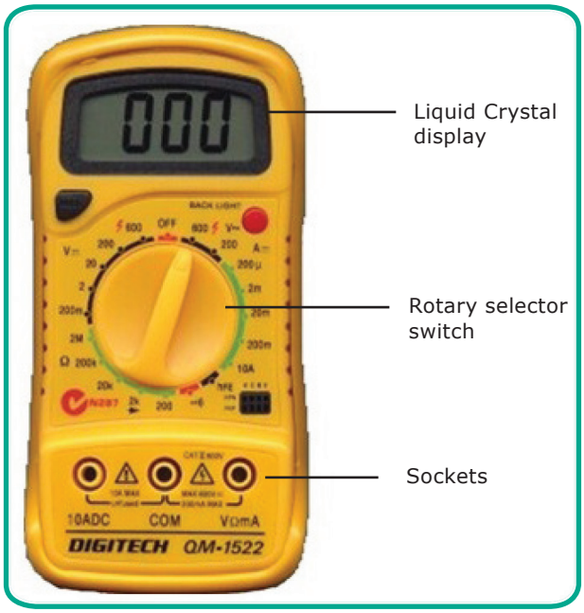


Figure 10.14 Digital Multimeter Front Panel

10.2.3.1. Components of a digital Multimeter

Figure 10.14 shows a typical digital multimeter. A digital multimeter comprises of three parts viz

1. Liquid Crystal Display
2. Rotary Selector switch and
3. Test probes.

The display usually has four digits and has the ability to display a negative sign. A few meters have illuminated displays for better viewing in low-light situations. The selector switch allows the user to set the multimeter to read different parameters such as current, voltage and resistance.

Two probes with banana plugs are plugged into two ports on the front of the instrument. COM stands for common and is always connected to ground or '-' of a circuit. VΩmA is the port in which the red probe (+ terminal) is conventionally plugged-in. This means, using this port, we can measure current up to 200 mA, voltage and resistance values.

10.2.3.2. Probe types

There are different types of probes available for multimeters. Here are some of the important probes in regular use.

1. Banana plugs to crocodile clips

   These cables are used for connecting large wires or pins on a bread board as shown in Figure 10.15. It is good for performing longer term tests where the user doesn't have to hold the probes in hands while testing the circuit.

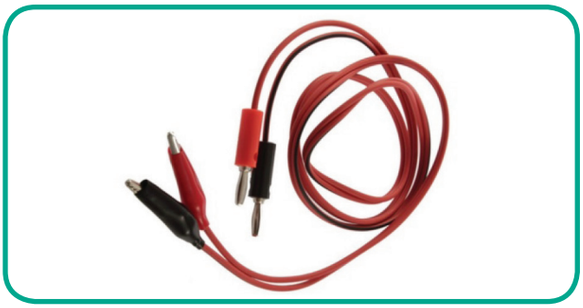


Figure 10.15 Banana Plugs to Crocodile Clips

2. Banana Plugs to IC Hook

   These probes are used to connect the legs of ICs with the meter and can be measured easily. It is shown in Fig 10.16

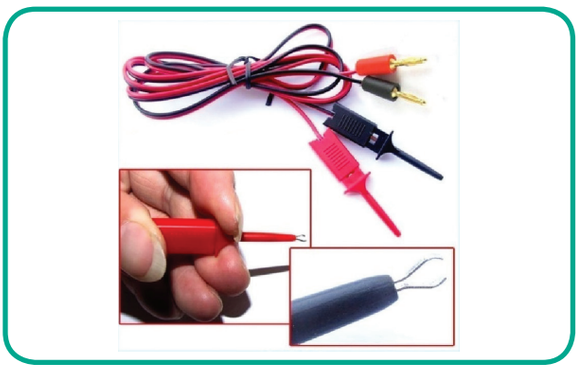


Figure 10.16 Banana Plugs to IC Hook

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